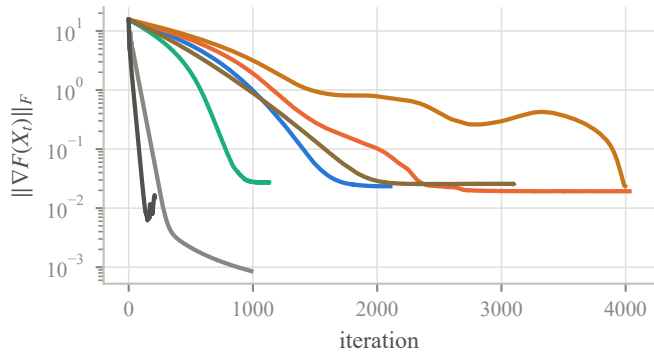
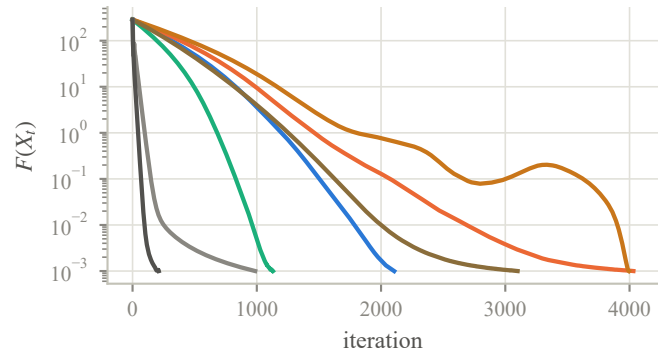


500x500, spectrum uniform, $L/\sigma = 1.65\text{e}+07$, lmo_dtype bfloat16



— SignMuon — EF21-MuonUSign — EF21-MuonSign — Muon — SignSGD — SGD — Adam